PRACTICAL SESSION 11

Task 1: Develop a multithreaded counter application using POSIX threads (`pthread_create`, `pthread_join`). Demonstrate race conditions by allowing multiple threads to update a shared variable concurrently.

```
  UW PICO 5.09                    File: counter_race.c


#include <stdio.h>
#include <pthread.h>

#define NUM_THREADS 4
#define ITERATIONS 1000000

long counter = 0;

void *increment(void *arg)
{
    for (int i = 0; i < ITERATIONS; i++)
    {
        counter++;
    }

    return NULL;
}

int main(void)
{
    pthread_t threads[NUM_THREADS];

    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_create(&threads[i], NULL, increment, NULL);
    }

    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_join(threads[i], NULL);
    }

    printf("Expected counter value: %d\n",
           NUM_THREADS * ITERATIONS);
    printf("Actual counter value: %ld\n", counter);

    return 0;
}
```

```
(base) saiabburi@Charishma-MacBook-Air ~ % mkdir -p ~/OS_Lab_Session11
(base) saiabburi@Charishma-MacBook-Air ~ % cd ~/OS_Lab_Session11
(base) saiabburi@Charishma-MacBook-Air OS_Lab_Session11 % clang --version
Apple clang version 14.0.3 (clang-1403.0.22.14.1)
Target: arm64-apple-darwin22.5.0
Thread model: posix
InstalledDir: /Library/Developer/CommandLineTools/usr/bin
(base) saiabburi@Charishma-MacBook-Air OS_Lab_Session11 % nano counter_race.c
```

```
(base) saiabburi@Charishma-MacBook-Air OS_Lab_Session11 % clang -Wall -Wextra -pthread counter_race.c -o counter_race
counter_race.c:10:23: warning: unused parameter 'arg' [-Wunused-parameter]
void *increment(void *arg)
                     ^
1 warning generated.
(base) saiabburi@Charishma-MacBook-Air OS_Lab_Session11 % ./counter_race
Expected counter value: 4000000
Actual counter value: 1116621
(base) saiabburi@Charishma-MacBook-Air OS_Lab_Session11 %
```

Task 2: Modify the previous program using mutex locks (`pthread_mutex_lock`, `pthread_mutex_unlock`) and compare the outputs with the race-condition version. Analyze the impact of synchronization.

UW PICO 5.09 File: counter_mutex.c

```c
#include <stdio.h>
#include <pthread.h>

#define NUM_THREADS 4
#define ITERATIONS 1000000

long counter = 0;
pthread_mutex_t lock = PTHREAD_MUTEX_INITIALIZER;

void *increment(void *arg)
{
    (void)arg;

    for (int i = 0; i < ITERATIONS; i++)
    {
        pthread_mutex_lock(&lock);
        counter++;
        pthread_mutex_unlock(&lock);
    }

    return NULL;
}

int main(void)
{
    pthread_t threads[NUM_THREADS];

    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_create(&threads[i], NULL, increment, NULL);
    }

    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_join(threads[i], NULL);
    }

    printf("Expected counter value: %d\n",
           NUM_THREADS * ITERATIONS);
    printf("Actual counter value: %ld\n", counter);

    pthread_mutex_destroy(&lock);

    return 0;
}
```

```
(base) saiabburi@Charishma-MacBook-Air OS_Lab_Session11 % clang -Wall -Wextra -pthread counter_mutex.c -o counter_mutex
(base) saiabburi@Charishma-MacBook-Air OS_Lab_Session11 % ./counter_mutex
Expected counter value: 4000000
Actual counter value: 4000000
(base) saiabburi@Charishma-MacBook-Air OS_Lab_Session11 %
```